

None



Preface

What Makes This Ebook Unique

Tips for Effective Use of This Ebook

[Author Biography](#)

[Other ebooks](#)

[Contacts and Social Media](#)

[Sources & References](#)

[Thank You!](#)

1st Edition

Why This E-book?

"The aim of this ebook is to give you the 'aha' moment right away at the start of learning a new concept."

- Practical step By Step Guide With Simple Examples
- Visual Illustrations and Interactive
- Simple Datasets
- Comprehensive Coverage (pandas Documentation used as reference)

#Import Libraries

import pandas as pd, numpy as np

pandas.isna(obj):

is used to detect missing

values in a DataFrame or

array-like objects e.g (Series).

Missing values are usually

represented as NaN (Not a Number)

in pandas. The isna function

returns a boolean DataFrame or

Series where the entries are

True if the corresponding entry

is NaN and False otherwise.

df:

	A	B	C
0	1.0	NaN	1.0
1	2.0	2.0	2.0
2	NaN	3.0	3.0
3	4.0	4.0	NaN



	A	B	C
0	False	True	False
1	False	False	False
2	True	False	False
3	False	False	True

Code Snippet:

pd.isna(df)

The "isna" syntax

pandas.isna(obj)

The `obj` parameter

The "obj" parameter accepts scalar or array-like values.
It is the only parameter for this function.

Scalar arguments (including strings)
result in a scalar boolean.

`pd.isna('dog') → False`

Scalar arguments (including strings)
result in a scalar boolean.

`pd.isna(pd.NA) → True`

Scalar arguments (including strings)

result in a scalar boolean.

`pd.isna(np.nan)` → True

ndarrays result in
an ndarray of boolean
s.

array:

```
array([[ 1., nan,  3.],
       [ 4.,  5., nan]])
```

output:

```
array([[False,  True, False],
       [False, False,  True]])
```

Code Snippet:

```
pd.isna(array)
```

Code Snippet:

```
array = np.array([[1, np.nan, 3], [4, 5, np.nan]])
array
```

array:

```
array([[ 1., nan,  3.],
       [ 4.,  5., nan]])
```

For Series and DataFrame, the
same type is returned, containing booleans.

df:

	A	B	C
0	1.0	NaN	1.0
1	2.0	2.0	2.0
2	NaN	3.0	3.0

→

	A	B	C
0	False	True	False
1	False	False	False
2	True	False	False

Code Snippet:

```
pd.isna(df)
```

	A	B	C		A	B	C
3	4.0	4.0	NaN	3	False	False	True

```
#try
#pd.isna(df[1])
```

For indexes, an ndarray of booleans is returned.

df: DatetimeIndex(
 ['2017-07-05', '2017-07-06',
 'NaT', '2017-07-08'],
 dtype='datetime64[ns]', freq=None)

→ array([False, False, True, False])

Code Snippet:
 pd.isna(index)

Tips for Effective Use of This Ebook

Use Examples First: If you don't understand the text, go straight to the examples—they're self-explanatory. After reviewing them, return to the text to grasp the practical concepts.

Practice with Datasets: Copy the datasets provided and practice using them to reinforce your understanding.

Sources & References

[pandas.crosstab Documentation](https://pandas.pydata.org/docs/reference/api/pandas.isna.html) (https://pandas.pydata.org/docs/reference/api/pandas.isna.html)

Author Biography

Kichere Magubu is a data enthusiast and content creator.

Other E-books

[Download](https://github.com/kicherethedatascientist/My_e-Books) (https://github.com/kicherethedatascientist/My_e-Books)

Contacts and Social Media

Kichere Magubu

 Dar es salaam, Tanzania

 [Kichere Magubu](#)

 [Kichere The Data Scientist](#)

 [KichereTheDataScientist](#)

 [KichereTheDataScientist](#)

✉ kicherethedatascientist@gmail.com

📷 kicherethedatascientist

📞 +255 654 729 851

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Thank you for reading this e-book! If you found it valuable, please consider leaving an honest review. Your feedback and support mean a lot to me!

Thank You!

```
#print("The cell to convert jupyter notebook to html")  
!jupyter nbconvert --to hide_code_html "pandas.isna.ipynb"
```